

Big Data Analytics — Portfolio Section 4

Task 4.2: Supply Chain Optimisation — Numerical Model Formulation

Three-Echelon Multi-Period Supply Chain Network | Profit Maximisation

1. Problem Overview

This model formulates a three-echelon, multi-period supply chain optimisation problem over a 12-month planning horizon. The network consists of 2 factories, 4 potential depots, and 10 retailers. The objective is to maximise total sales profit by determining optimal production quantities, depot selections, and product transportation routes.

Products may flow via two valid paths:

- Factory → Depot → Retailer
- Factory → Retailer (direct shipment)

Depots act as transshipment nodes only — they cannot store inventory. Factories can hold inventory and may experience shortages, both of which incur costs.

2. Sets

Notation	Set Name	Elements
$F = \{1, \dots, f\}$	Factory set	Swindon, Sheffield
$D = \{1, \dots, d\}$	Depot set	Leicester, Coventry, Bristol, Bedford
$R = \{1, \dots, r\}$	Retailer set	Nottingham, Cambridge, Cardiff, Norwich, Portsmouth, Birmingham, Exeter, Preston, Leeds, Oxford
$T = \{1, \dots, t\}$	Period set	Months 1 to 12

3. Parameters

Symbol	Description	Value / Source
$C1_{fd}$	Unit transport cost: factory f to depot d	See Table 3 (£/unit)
$C2_{fr}$	Unit transport cost: factory f to retailer r	See Table 3 (£/unit)
$C3_{dr}$	Unit transport cost: depot d to retailer r	See Table 3 (£/unit)
C_f	Unit production cost at factory f	£225 per unit (both factories)
C_d	Operating cost of depot d per period	Leicester £16,000 Coventry £13,000 Bristol £12,500 Bedford £14,000

H	Unit holding cost (inventory)	£20 per unit per month
S	Unit shortage cost	£200 per unit per month
P _{rt}	Sales price to retailer r in period t	Forecasted from sales_price_2years.csv (Task 4.1)
MS _f	Max production capacity of factory f	Swindon: 1,000 units/month Sheffield: 1,250 units/month
MT _d	Max transshipment capacity of depot d	Leicester: 300 Coventry: 200 Bristol: 300 Bedford: 150
B _{rt}	Demand at retailer r in period t	Forecasted from retailer_demand_3years.csv (Task 4.1)
I0 _f	Initial inventory at factory f	Swindon: 0 units Sheffield: 100 units

4. Decision Variables

Variable	Description	Domain
x1 _{fdt}	Units shipped from factory f to depot d in period t	$x1_{fdt} \geq 0$
x2 _{frt}	Units shipped from factory f directly to retailer r in period t	$x2_{frt} \geq 0$
x3 _{drt}	Units shipped from depot d to retailer r in period t	$x3_{drt} \geq 0$
Prod _{ft}	Production quantity at factory f in period t	$Prod_{ft} \geq 0$
Inven _{ft}	Inventory at factory f at end of period t	$Inven_{ft} \geq 0$
Short _{ft}	Shortage at factory f at end of period t	$Short_{ft} \geq 0$
y _d	Binary: 1 if depot d is open; 0 if closed	$y_d \in \{0, 1\}$
z _{nr}	Binary: 1 if node n supplies retailer r; 0 otherwise (Task 4.4)	$z_{nr} \in \{0, 1\}$

5. Objective Function

Maximise total profit over the 12-month planning horizon:

$$\text{Maximise } Z = \text{Revenue} - \text{Production Cost} - \text{Transport Cost} - \text{Depot Cost} - \text{Holding Cost} - \text{Shortage Cost}$$

Expanded form using Σ summation notation:

$$\begin{aligned}
 Z = & \sum(r \in R, t \in T) [P_{rt} \times (\sum(f \in F) x2_{frt} + \sum(d \in D) \\
 & x3_{drt})] \\
 & - \sum(f \in F, t \in T) [C_f \times Prod_{ft}] \\
 & - \sum(f \in F, d \in D, t \in T) [C1_{fd} \times x1_{fdt}] \\
 & - \sum(f \in F, r \in R, t \in T) [C2_{fr} \times x2_{frt}] \\
 & - \sum(d \in D, r \in R, t \in T) [C3_{dr} \times x3_{drt}] \\
 & - 12 \times \sum(d \in D) [C_d \times y_d] \\
 & - \sum(f \in F, t \in T) [H \times Inven_{ft}] \\
 & - \sum(f \in F, t \in T) [S \times Short_{ft}]
 \end{aligned}$$

Where:

- Line 1: Total revenue — price $\times$ units delivered to all retailers via all routes
- Line 2: Total production cost (£225 per unit at both factories)
- Lines 3–5: Total transport cost across all three route types
- Line 6: Depot operating cost — C_d is defined as cost per period (Table 4), $\times$ 12 monthly periods
- Line 7: Inventory holding cost (£20 per unit per month at factories)
- Line 8: Shortage penalty cost (£200 per unit per month at factories)

6. Constraints

Constraint 1 — Factory Production Capacity

Production at each factory in each period cannot exceed its maximum monthly capacity:

$$\text{Prod}_{ft} \leq \text{MS}_f \quad \forall f \in F, \quad t \in T$$

Constraint 2 — Depot Transshipment Capacity

Total flow through a depot cannot exceed its capacity. If $y_d = 0$ (depot closed), no flow is permitted:

$$\sum (f \in F) x_{1_f d t} \leq \text{MT}_d \times y_d \quad \forall d \in D, \quad t \in T$$

Constraint 3 — Depot Flow Balance (Transshipment Only)

Depots cannot store inventory. All goods received must be dispatched within the same period:

$$\sum (f \in F) x_{1_f d t} = \sum (r \in R) x_{3_d r t} \quad \forall d \in D, \quad t \in T$$

Constraint 4 — Retailer Demand Satisfaction

Each retailer must receive exactly their forecasted demand every period. Demand must be fully met:

$$\sum (f \in F) x_{2_f r t} + \sum (d \in D) x_{3_d r t} = B_{rt} \quad \forall r \in R, \quad t \in T$$

Constraint 5 — Factory Inventory / Shortage Balance

The inventory or shortage at each factory is updated each period as follows:

$$\text{Inven}_{f,t-1} - \text{Short}_{f,t-1} + \text{Prod}_{ft} - (\sum_d x_{1_f d t} + \sum_r x_{2_f r t}) = \text{Inven}_{ft} - \text{Short}_{ft}$$

For $t = 1$, the opening inventory is $I0_f$ (from Table 1) and opening shortage is 0. This ensures inventory and shortages carry forward, production adds to supply, and shipments reduce supply.

Non-Negativity and Binary Constraints

$$\begin{aligned} x_{1_f d t}, \quad x_{2_f r t}, \quad x_{3_d r t}, \quad \text{Prod}_{ft}, \quad \text{Inven}_{ft}, \quad \text{Short}_{ft} &\geq 0 \\ y_d &\in \{0, 1\} \quad \forall d \in D \end{aligned}$$

7. Additional Constraints (Task 4.4)

The following three logical constraints are added simultaneously to the base model. Additional binary variables z_{nr} are introduced ($= 1$ if node n supplies retailer r in any period; 0 otherwise), linked to flow variables using Big-M constraints ($M = 100,000$).

Additional Constraint i — Leicester Cannot Supply Both Nottingham and Portsmouth

Leicester depot may supply Nottingham or Portsmouth, but not both simultaneously:

$$z_{\text{Leicester, Nottingham}} + z_{\text{Leicester, Portsmouth}} \leq 1$$

Additional Constraint ii — Birmingham Supplier Must Also Supply Cardiff

If any node n supplies Birmingham, that same node must also supply Cardiff:

$$z_{n, \text{Birmingham}} \leq z_{n, \text{Cardiff}} \quad \forall n \in F \cup D$$

Additional Constraint iii — Cardiff Single Supplier

Cardiff can be supplied by exactly one factory or depot:

$$\sum_{n \in F \cup D} z_{n, \text{Cardiff}} \leq 1$$

Linking Constraints for z Variables (Big-M Method)

The binary supply indicator z_{nr} is linked to flow variables using Big-M constraints:

$$\text{Total_flow}(n \rightarrow r) \leq M \times z_{nr} \quad (z_{nr} = 0 \text{ forces flow to zero})$$

$$\text{Total_flow}(n \rightarrow r) \geq z_{nr} \quad (z_{nr} = 1 \text{ requires at least 1 unit of flow})$$

$$z_{nr} \in \{0, 1\} \quad \forall n \in F \cup D, \quad r \in R$$

Note: Predicted demand B_{rt} and price P_{rt} values are derived from Task 4.1 forecasting (Linear Regression and SARIMA models).